

INFR11287 Advanced Topics in Natural Language Processing

Mock Examination Paper 3: Mark Scheme

This mark scheme gives indicative points. Award credit for equivalent answers that demonstrate the same understanding.

1. Vision-language assistant. Total: 15 marks.

- (a) The task requires connecting language to visual objects, regions, text in images, and situated context. It addresses grounding because meanings of words such as “this”, labels, artefacts, and spatial relations depend on perception rather than text alone. Award up to 3 marks.
- (b) Components: vision encoder converts image/patches into visual features; projector/connector maps visual features into the LLM hidden/token space; backbone LLM conditions on visual tokens and text to generate answers. Award up to 4 marks.
- (c) Linear connector: cheap and stable but less expressive. MLP: more expressive with moderate cost and possible overfitting/initialisation issues. Attention-based connector: can select salient visual information but is costlier and harder to train. Award up to 3 marks.
- (d) Data: exhibit images with captions, OCR labels, VQA pairs, referring expressions, multilingual labels, accessibility-oriented egocentric photos, expert-curated museum facts. Risks: licensing, privacy of visitors, cultural misrepresentation, hallucinated labels, domain shift from generic datasets. Award up to 3 marks.
- (e) Intrinsic: captioning, OCR, VQA, object recognition, hallucination benchmarks. Extrinsic: visitor task success, answer usefulness, accessibility studies with visually impaired users, latency. Ethics/accessibility: wrong descriptions, cultural harms, privacy, overconfident unsafe guidance. Award up to 2 marks.

2. Agents and computer use. Total: 15 marks.

- (a) An agent perceives an environment through sensors and acts through actuators. API design: observations are tool outputs/form states and actions are API calls. Browser design: observations are screenshots/OCR/accessibility trees and actions are clicks, typing, scrolling, navigation, and stopping. Award up to 3 marks.
- (b) Text-only/API agent: more reliable and structured when APIs exist, easier validation; disadvantage is limited to available tools and schemas. Vision-based CUA: can operate across arbitrary GUIs; disadvantage is visual grounding, UI fragility, and action risk. Award up to 4 marks.
- (c) I is the instruction/goal; s_t is the environment state at step t , often a screenshot; a_t is the action such as click/type/scroll/terminate. History matters for tracking progress, avoiding repeated actions, recovering from mistakes, and interpreting state changes. Award up to 3 marks.
- (d) Failure modes: clicking wrong button, stale UI after page changes, entering private data in wrong field, loops, irreversible cancellation/payment, prompt injection in webpage text. Mitigations: constrained action space, confirmations, sandboxing, API preference, state checks, logging, time/action limits, human approval for irreversible actions. Award up to 3 marks.
- (e) Final success alone hides path quality and risk. Additional measures: intermediate progress, visual grounding accuracy, loop rate, number of actions, recovery/backtracking, unsafe action rate, latency, robustness to UI changes. Award up to 2 marks.

3. MCQ answer key. Total: 20 marks.

- (a) B
- (b) C
- (c) A

- (d) A
- (e) A
- (f) A
- (g) B
- (h) A
- (i) A
- (j) C